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Comparative Characterization of a Proposed Generic Nusinersen: Identity of the Oligonucleotide Structure and Equivalence in SMN2 Splicing Activity

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Abstract

Background/Objectives: Nusinersen is a synthetic antisense RNA oligonucleotide employed in the management of spinal muscular atrophy, a rare neuromuscular disorder, by modulating the alternative splicing of the survival motor neuron 2 (SMN2) gene. GNR-100 represents the first generic version of the reference listed drug (RLD), containing nusinersen sodium as the active pharmaceutical ingredient. We performed comprehen-sive evaluations in accordance with FDA guidelines, including side-by-side comparative analyses of critical quality attributes, to thoroughly characterize the structural and func-tional properties of both nusinersen products. **Results/Methods**: GNR-100 was compre-hensively demonstrated to be highly similar to RLD in terms of oligonucleotide structure, physicochemical properties, impurity profile, and in vitro cell-based assays for SMN-gene splice-switching and SMN-protein activity. Structural analyses confirmed that the oligo-nucleotide primary sequences and chemical structures were identical. The diastereomeric composition and higher-order structures were also similar between the proposed generic and the reference product. Comparable resistance to phosphodiesterase degradation and nearly identical melting temperatures of the oligonucleotide duplexes with their comple-mentary strand further substantiated the structural sameness of the nusinersen products. The impurity profile of the proposed therapeutic oligonucleotide was consistent with that of RLD, and the collectively reduced levels of impurities, as assessed by orthogonal ana-lytical methods, indicated no meaningful impact on the safety profile. Moreover, both products exhibited comparable biological activity in enhancing the production of full-length SMN2 mRNA transcripts and functional SMN protein in fibroblasts derived from SMA patients. **Conclusions**: These quality studies demonstrate that GNR-100 exhibits no significant differences from the licensed drug across structural, physicochemical, biophys-ical, and biological attributes, establishing its potential as a cost-effective therapeutic al-ternative for patients with spinal muscular atrophy.

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1. Introduction

Nusinersen is an 18-mer synthetic antisense RNA oligonucleotide specifically de-signed to manage spinal muscular atrophy (SMA), a rare autosomal recessive neuromus-cular disease characterized by degeneration of the motor neurons in the anterior horn of

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the spinal cord, resulting in atrophy of the voluntary muscles of the limbs and trunk [1]. The disease is caused by deletions or mutations in the survival motor neuron 1 gene (SMN1) on the long arm of chromosome 5 (5q13), which results in the absence of SMN protein production [2,3]. This deficit can be partially compensated for by the expression of a nearly identical SMN2 gene. However, due to the single nucleotide C > T difference, skipping of exon 7 occurs in most SMN2 mRNA transcripts, resulting in the encoding of an SMN protein isoform that is unstable and does not function in the same manner as the full-length SMN isoform [4,5]. Nusinersen increases the proportion of full-length SMN2 exon 7 mRNA transcripts by binding to the intronic splicing silencer (ISS N1) located in intron 7 of the SMN2 mRNA precursor. After SMN2 mRNA is synthesized, it can be trans-lated into SMN protein with a full polypeptide chain length and normal functional activity [6,7]. To increase the stability of nusinersen during use, ribose residues are modified with 2′-O-2-methoxyethyl moieties, while the phosphodiesters are replaced with phos-phorothioate linkages [8]. Nusinersen is administered directly to the central nervous sys-tem via intrathecal injection, with nusinersen sodium as the active pharmaceutical ingre-dient. Following distribution to the CNS and peripheral tissues, the drug is metabolized via slow hydrolysis by exonucleases, with the primary route of elimination of nusinersen and its metabolites likely being urinary excretion [1,9,10].

Generic formulations of original drugs with proven long-term clinical efficacy can provide equivalent clinical effects at a reduced cost to society [11,12]. GNR-100 is the first generic version comprising nusinersen sodium as the active pharmaceutical ingredient. GNR-100 was approved in Russia to provide greater patient access to nusinersen with efficacy and safety comparable to the marketed reference product. It should be noted that an active ingredient such as nusinersen is classified as "complex generic", for which reg-ulatory approaches to confirm bioequivalence to the reference drug are still under devel-opment and require more attention and consideration of possible risks, thereby imposing additional responsibility on the developer. We conducted comprehensive studies based on FDA guidelines [13], including side-by-side comparative analyses of critical quality attributes, to characterize the structural and functional properties of the proposed and reference nusinersen products synthesized by the conventional automated phospho-ramidite method [14,15]. The proposed and reference products were investigated in terms of oligonucleotide structure, physicochemical properties, impurity profile, and in vitro cell-based assays for SMN-gene splice-switching and SMN-protein activity. The quality attributes of GNR-100 were assessed against the reference nusinersen using a detailed characterization described herein to ensure that the functionality and safety of the pro-posed product are comparable to those of the reference product.

2. Materials and Methods

2.1. Production and Purification of Nusinersen

GNR-100 was synthesized via the standard solid-phase method with 2′-MOE phos-phoramidites as the starting material (Figure S1) using well-established automated chem-istry [14,15]. The phosphoramidites used to synthesize nusinersen were homogeneous in their chirality and represent commercially available nucleoside precursors. Supplier qual-ity was confirmed by rigorous vendor qualification and incoming quality control. The structure and purity of each phosphoramidite were verified by mass spectrometry, $_{1}$H and $_{31}$P NMR spectroscopy, and HPLC-RP. The stereochemical integrity of the phospho-ramidites remained unchanged throughout the entire manufacturing process with no ob-served racemization.

The optical density of the cleaved DMT ion was monitored in real time during each synthesis cycle to enable precise control over coupling efficiency and overall process per-formance, allowing for the identification and mitigation of incomplete or failed sequences.

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To ensure structural and qualitative equivalence of GNR-100 to RLD, a tailored purification strategy was developed, employing hydrophobic and ion-exchange HPLC to se-lectively remove process- and product-related impurities generated during synthesis. A comprehensive analytical control strategy was implemented throughout the process, in-corporating ion-pair high-performance liquid chromatography-mass spectrometry (HPLC-MS) for the precise characterization of both intermediates and the final nusinersen product.

2.2. Study Design

Comparative analyses were conducted using 7 production batches of the generic drug product manufactured within a year and 6 commercial batches of the RLD sourced from the European Union, spanning varying expiration dates. The study design, quality attributes, and analytical procedures were aligned with the U.S. Food and Drug Administration guidance for development of generic nusinersen [13]; the regulatory requirements of the local authority in the Eurasian Economic Union (EAEU); and established sci-entific literature on therapeutic oligonucleotide development, including guidelines from the European Pharmaceutical Oligonucleotide Consortium (EPOC) (http://www.epoc.dev (accessed on 15 November 2025).

Analytical characterization was conducted as detailed in Appendix A.1.

Primary fibroblast samples were derived from three donors diagnosed with spinal muscular atrophy.

2.3. Statistical Evaluation

A quality range (QR) approach was employed to evaluate the comparability of GNR-100 to the RLD product. The QR limits were defined based on the variation observed in the reference product, calculated as the mean ± 2 (or 3) standard deviation (SD). Compa-rability was established if at least 90% of the values from generic (GNR-100) batches fell within the quality range derived from multiple innovator reference product batches [16,17]. In certain cases, a direct visual comparison was also performed by superimposing the spectral or other profiles of the test product onto the corresponding profiles from the reference batches.

To estimate specific activity, the measured activity for each sample was expressed as a percentage of a reference product lot, which was designated as the standard activity value of 100%.

3. Results

3.1. Physicochemical and Biological Characterization

Current analytical procedures were employed to conduct a comprehensive structural and functional characterization of GNR-100 and the reference nusinersen. The analysis elucidated the primary sequence, chemical structure, and diastereomeric composition alongside spectral signatures, molecular mass, oligonucleotide content, related sub-stances, and key biological attributes of the compared drug products. Where feasible, comparisons were performed in a single head-to-head analytical run under identical con-ditions across all samples, as this approach enhances the precision and reliability of de-tecting subtle differences or similarities. Summary results of the comparative characteri-zation of nusinersen drug products are presented in Table 1.

Table 1. Comparative characterization of nusinersen drug products.

Parameter	Assay	CQA	QR (RLD)	GNR-100
Primary structure	LC-MS	Monoisotopic	6 ions	6 ions

			Z(−3−8)	Z(−3−8)
	LC-MS/MS	Nucleotide sequence	matches with theoretical	matches with theoretical
		^{1}H NMR spectrum	spectrum profile	confirmed
	NMR spectroscopy	^{31}P NMR	spectrum profile	confirmed

^{31}P NMR (R_P/S_P)		0.74 ± 0.03		0.82 ± 0.05
^{31}P NMR (P_O/P_S)		≤0.005		<0.001
^{13}C NMR spectrum		spectrum profile		confirmed
Chemical structure		2D NMR ^{1}H-^{13}C	spectrum profile	confirmed
and diastereomeric	CD spectroscopy		spectrum profile	confirmed
state	FTIR spectroscopy	3D chemical structure	spectrum profile	confirmed
	IF spectroscopy		spectrum profile	confirmed

Nuclease resistance

PDE I treatment, %			sustained	sustained
PDE II treatment, %			sustained	sustained
	Complementary duplex melting point	Tm, °C	61.5 ± 0.2	62.0 ± 0.2
Molecular mass	LC-MS	Monoisotopic mass, Da	7122.24 ± 0.01	7122.24 ± 0.01
	AUC	Molecular mass, kDa	5.98 ± 0.09	6.01 ± 0.13
In vitroactivity	Cell-based assay	SMN production, %	75.1–125.8	83.3–95.7
	RT PCR	SMN mRNA expression, %	87.8–111.3	90.0–96.0
	SEC HPLC	Monomer, %	99.94–99.97	100
		Aggregates, %	0.03–0.06	n.d.
	Ion-pairing RP HPLC			

Main substance, %		97.2–99.6	98.8 ± 0.1	
Related, %		0.4–2.8	1.2 ± 0.1	
		Nusinersen, %	92.7–93.9	95.7–96.4
		Oligonucleotides (PO),	1.59–1.86	0.73–1.52
		Oligonucleotides (n − 1) [a]	0.84–1.55	0.51–0.59
		Oligonucleotides (n + 1) [a]	0.37–0.63	0.48–0.52
		Oligonucleotides (n − 2) [a]	0.01–0.03	n.d.
		Abasic site, %	0.22–0.27	0.10–0.17
		CNET, %	0.24–0.28	n.d.
		ADP, %	0.02–0.03	n.d.
	Ion-pairing RP LC-MS	IDP, %	0.29–0.33	0.28–0.31
Impurities and related substances		2′-OMe, %	0.86–0.95	0.36–0.46
		MAM, %	n.d.	n.d.
		2′-O-(2-ethoxyethyl)/ dithioate, %	0.45–0.90	0.23–0.29
		DMT, %	n.d.	n.d.
		AMPA, %	0.05–0.24	n.d.
		Dimer, %	0.24–0.27	0.31–0.32
		Unidentified, %	0.38–0.52	0.3 ± 0.1
	ICP-OES	Na, mg/L	–	50 ± 20
		B	Does not exceed the permitted daily exposure (PDE) for each element in accordance with ICH guideline Q3D (R2) on elemental impurities	
		Fe		
	ICP-MS	P		
		S		
		Si		

3.2. Evaluation of Chemical Structure and Diastereomeric Composition

In all instances, the measured monoisotopic masses of nusinersen (M) and the corresponding mass-to-charge ratios (m/z) of its deprotonated ions deviated from the

theoretical values by no more than 6 ppm, indicating structural similarity among the stud-ied drug products (Table 2, Figure 1). The molecular masses of the compared drug prod-ucts were consistent within the analytical uncertainty of the high-speed analytical ultra-centrifugation method. However, given the inherent structural complexity of oligonucle-otides, molecular mass determination—despite their high-resolution—proved insufficient to conclusively establish structural identity between GNR-100 and the reference nusinersen.

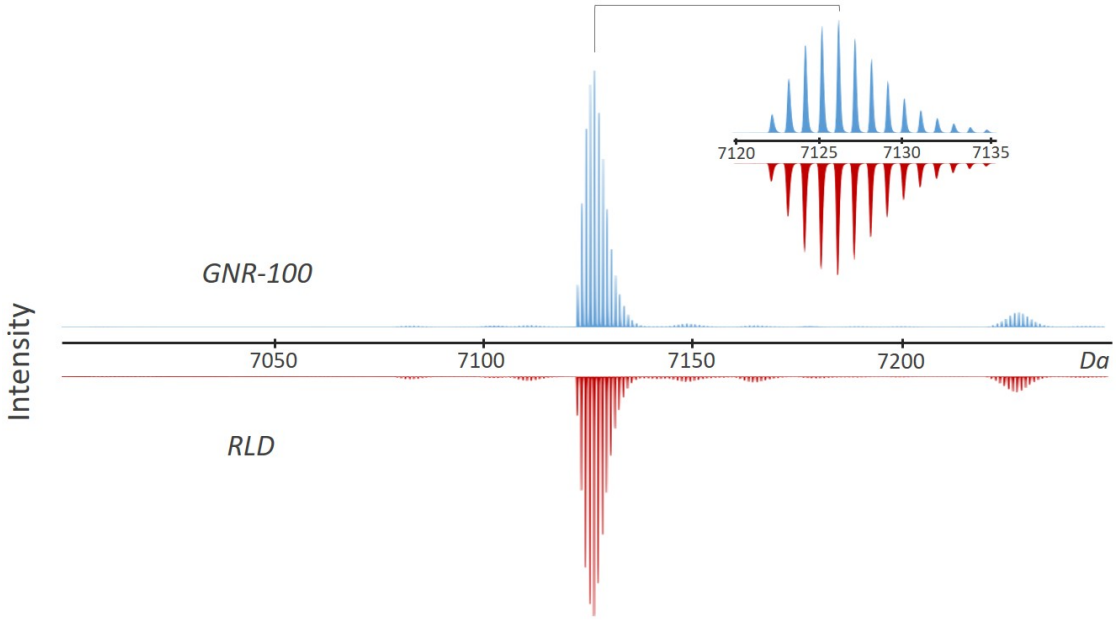

Figure 1. The mass spectrum of nusinersen. A detailed isotopic distribution is shown within an inset.

Table 2. Mass-to-charge ratio for the nusinersen deprotonated ions.

Ion	$(M-8H)_{8-}$	$(M-7H)_{7-}$	$(M-6H)_{6-}$	$(M-5H)_{5-}$	$(M-4H)_{4-}$	$(M-3H)_{3-}$
m/z (theo.)	889.28	1016.46	1186.04	1423.45	1779.56	2373.09
GNR-100	889.27	1016.46	1186.04	1423.45	1779.56	2373.08
Δ, ppm	3.7	3.6	2.2	1.9	1.1	0.3
RLD	889.27	1016.46	1186.04	1423.45	1779.56	2373.09
Δ, ppm	4.9	4.3	2.8	1.4	0.9	0.4

To further elucidate structural equivalence, collision-induced dissociation (CID) was employed to analyze the fragmentation pattern of the isolated ion, $(M-7H)_{7-}$ (m/z 1016.461), yielding comprehensive fragment ion spectra that fully span the oligonucleo-tides primary sequence (Table S1). Comparative analysis of these fragmentation profiles demonstrated a high degree of similarity between the drug products (Figure 2).

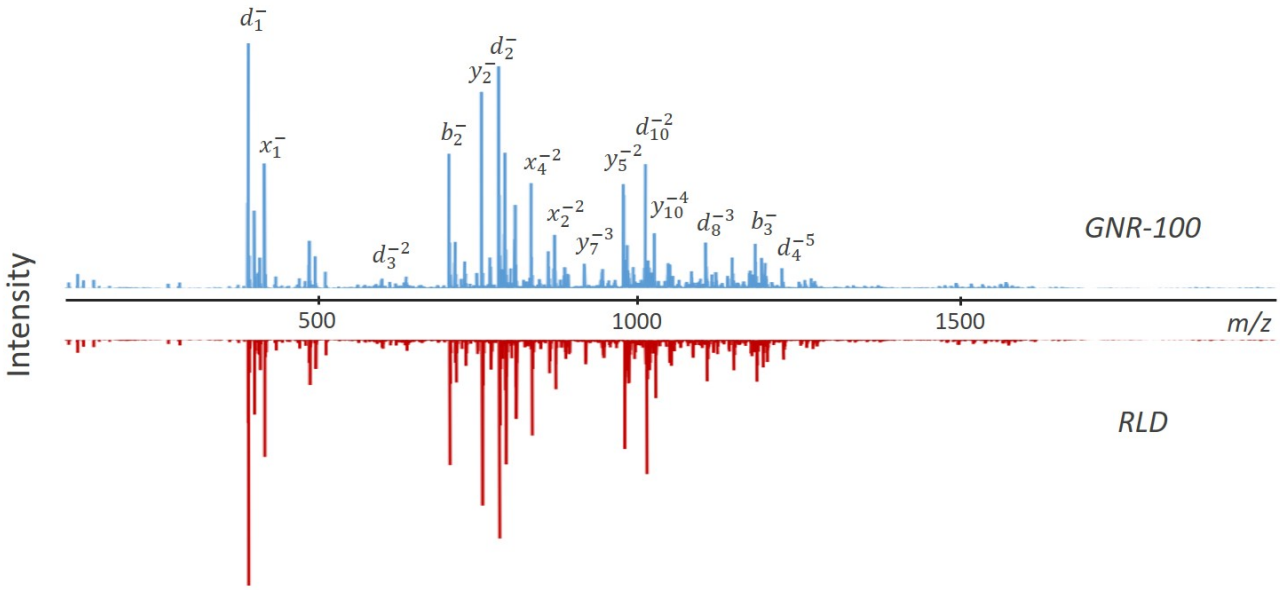

Figure 2. Tandem mass spectrum for $(M\text{-}7H)_{7-}$ ion. Basic fragment ions are marked with letters.

The $_1$H NMR spectra of the compared drug products exhibited highly similar signal patterns (Figure 3). Resonances in the δH range of 1.6–2.2 ppm were assigned to the methyl protons of thymine (Thy H_7) and 5-methylcytosine ($_{Me}$Cyt H_7). In the δH region of 3.0–4.5 ppm, signals of the methoxyethyl group protons (MOE $H_{2,3}$ and H_1) were overlapped with those of the ribose ring protons (R $H_{2,3,4,5}$). Notably, the ribose proton at position 1′ (R H_1) resonated downfield at 5.5–6.3 ppm, distinct from the other sugar protons. The resonances of the nitrogenous base protons were observed in the weak field (δH 7.1–8.8 ppm) and are typical for positions 2 (Ade H_2), 6 (Thy H_6, $_{Me}$Cyt H_6), and 8 (Ade H_8, Gua H_8). Due to acquisition in fluid-attenuated mode, certain individual proton signals were not resolved. Integration of the $_1$H spectral peaks corroborated the anticipated nucleotide composition and confirmed the structural equivalence of the nitrogenous base proton en-vironments between the compared drug products (Figure S2).

Figure 4 represents the $_{31}$P NMR spectra of GNR-100 and reference nusinersen, revealing nearly identical signal profiles in both samples. Two distinct, intense clusters, cor-responding to diastereomeric phosphorothioate (*PS*) linkages, appear in the range of 53–60 ppm [18]. Chemical shift referencing was accomplished using the prominent resonance of free phosphate (*Pi*).

The $_{13}$C NMR spectra of all samples exhibit a highly similar signal profile (Figure 5). Methyl carbons of thymine (Thy C_7) and 5-methylcytosine ($_{Me}$Cyt C_7) resonate in the range of 11–14 ppm. A distinct signal from the methoxyethyl carbon (MOE C_1) appears at 58 ppm, while the C_2 and C_3 carbons of the MOE-group are shifted downfield to 71 ppm and 69 ppm, respectively. The following signals of ribose residues are observed in the imme-diate vicinity at 62–67 ppm (R C_5), 72–74 ppm (R C_3), 79–85 ppm (R $C_{2,4}$), and 85–90 ppm (R C_1). The majority of nitrogenous base carbon signals are positioned in the downfield region. Four distinct resonances corresponding to the C_5 atoms ($_{Me}$Cyt, Thy, Gua, and Ade) fall within the 104–119 ppm of δC range. Two prominent clusters appear at the extreme downfield: one at 130–140 ppm and the other at 148–160 ppm. The first cluster corre-sponds to C_6 and C_8 ($_{Me}$Cyt C_6, Thy C_6, Gua C_8, Ade C_8). The second cluster arises from the resonance of atoms in positions 2, 4, and 6 ($_{Me}$Cyt $C_{2,4}$, Thy C_2, Gua $C_{2,4,6}$, Ade $C_{2,4,6}$). The C4 carbon of the thymine (Thy C_4) exhibits a unique, isolated peak at 165 –167 ppm.

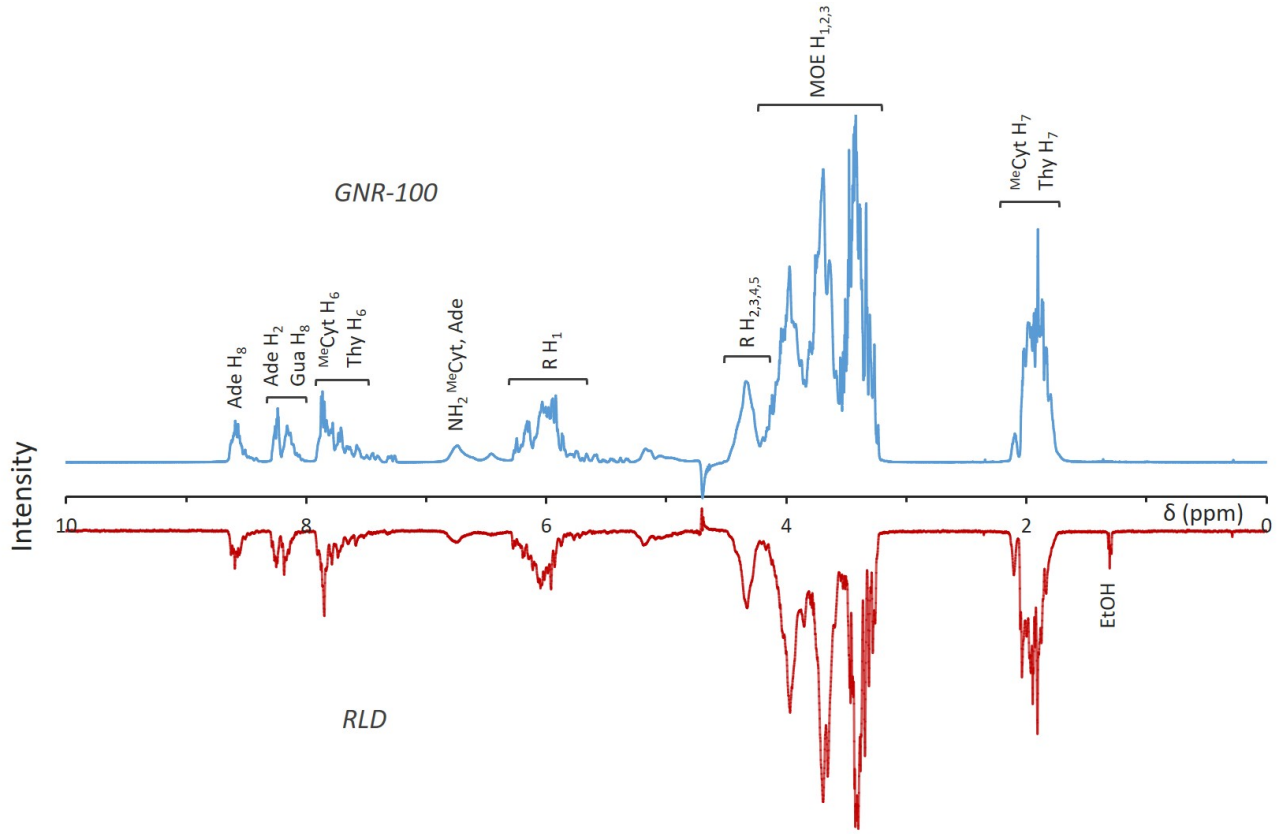

Figure 3. An example of nusinersen $_{1}$H NMR spectrum.

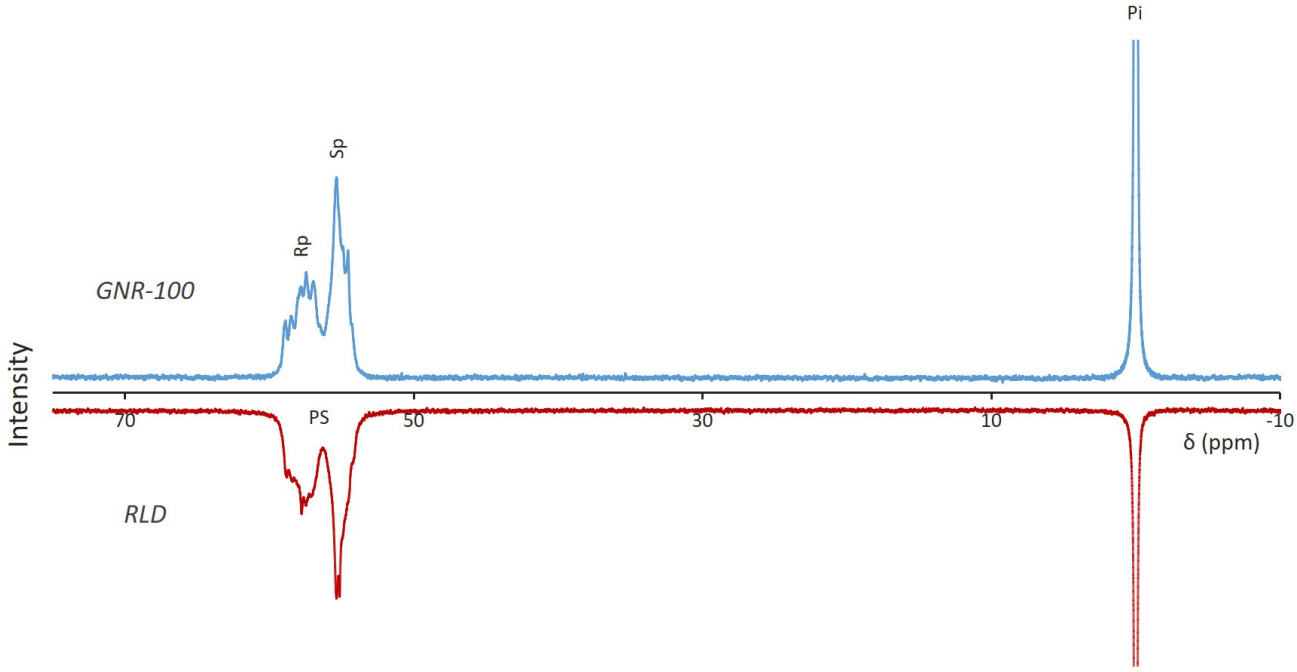

Figure 4. An example of nusinersen $_{31}$P NMR spectrum.

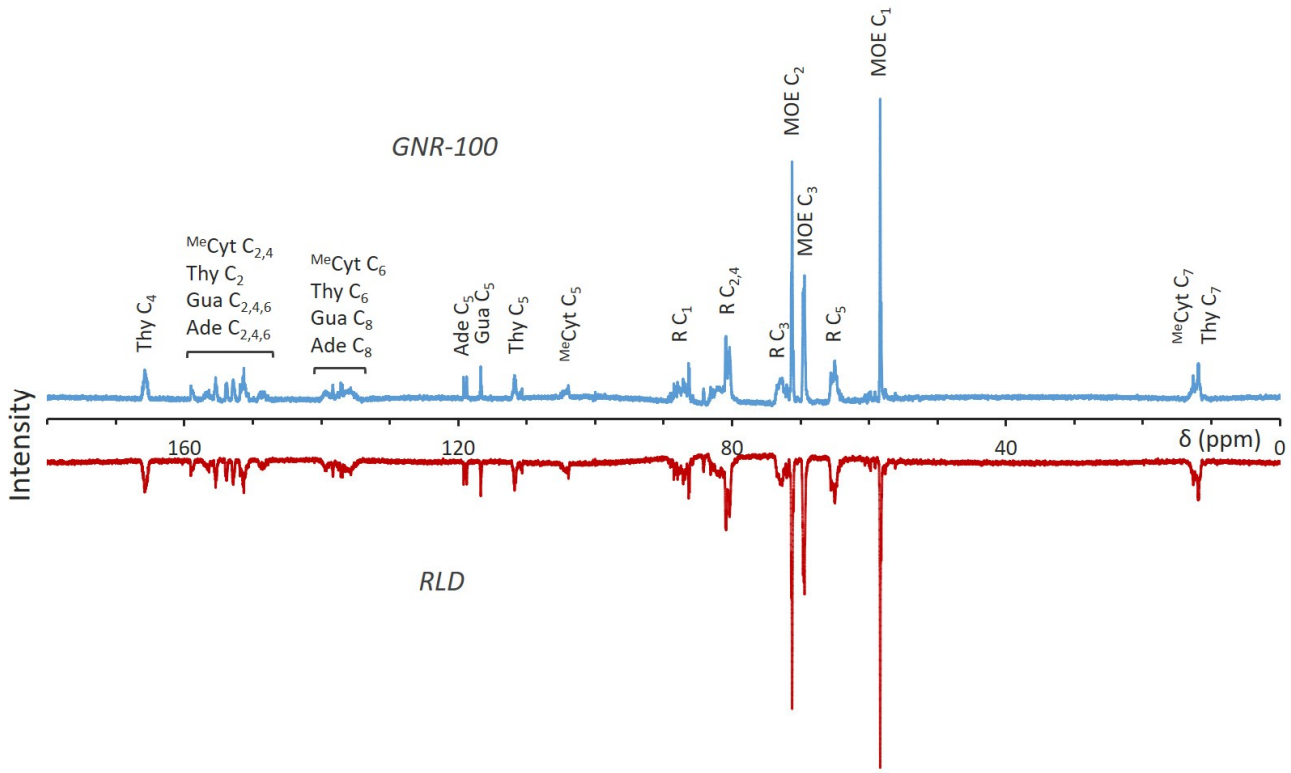

Figure 5. An example of nusinersen $_{13}$C NMR spectrum.

Figure S3 demonstrates the two-dimensional $_{1}$H-$_{13}$C HSQC NMR spectra of GNR-100 and reference nusinersen. The region of aliphatic CH groups (δC/δH 55–90/2.5–6.5 ppm) was characterized by the highest signal density, where the resonances of the methoxyethyl (MOE) and ribose (R) groups appeared. The cross peaks corresponding to thymine (Thy C_7) and/or 5-methylcytosine ($_{Me}$Cyt C_7) methyl groups were distinct at δC/δH 12–13/1.7–2.1 ppm. The resonance of CH groups caused lower-intensity signals in the region of aro-matic atoms (δC/δH 134–155/7.2–8.8 ppm) ($_{Me}$Cyt C_6, Thy C_6, Gua C_8, and Ade $C_{2,8}$). The two-dimensional spectra align well with the one-dimensional $_{1}$H and $_{13}$C NMR spectra and the theoretical structure of nusinersen. Figure S4 shows the TOCSY NMR spectrum of GNR-100. Notably, all spectra exhibited highly overlapping signals attributable to the molecules heterogeneous spatial structure, particularly the distribution of phosphorothi-oate stereoisomers.

To further confirm the structural identity of GNR-100 and the reference nusinersen, both were evaluated for their resistance to broad-spectrum phosphodiesterases: Snake Venom Phosphodiesterase (PDE I) and Bovine Spleen Phosphodiesterase (PDE II), which have corresponding human homologs [19]. An unmodified control (CTRL) oligonucleo-tide, containing exclusively phosphodiester nucleotide linkages, served as a positive con-trol. Following incubation at 37°C for one hour, both the reference listed drug (RLD) and GNR-100 demonstrated marked resistance to enzymatic degradation (Figure S5). The sig-nal intensity of some reference nusinersen samples decreased by 10–15%, which corre-sponds to the error magnitude. However, treatment of GNR-100 samples did not change the intensity of the observed bands. In contrast, the control oligonucleotide underwent complete degradation within the same time frame.

Comparative melting point temperature (Tm) measurements of nusinersen duplexes with the complementary oligonucleotide ANT revealed negligible variation within 0.5 °C. The thermal transition points of GNR-100 closely mirrored those of the reference

nusinersen (Figure 6). In combination with NMR findings, this confirms the absence of meaningful differences in diastereomeric composition [20].

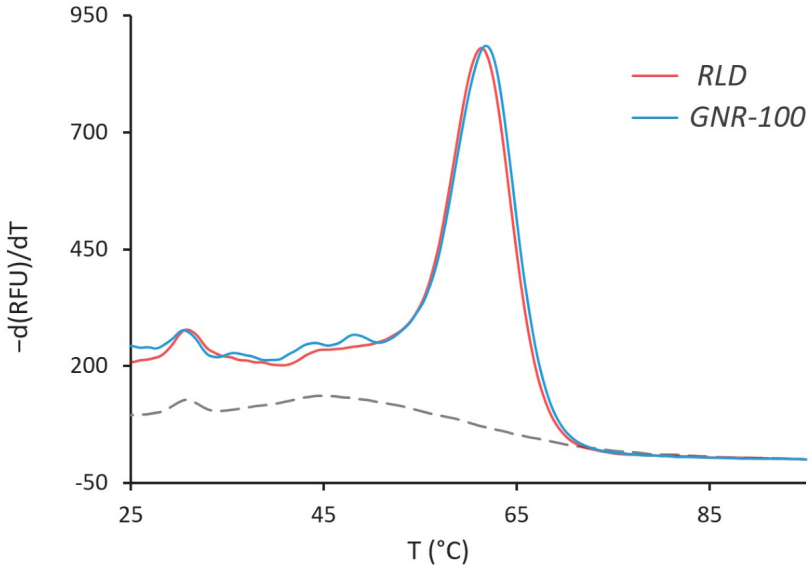

Figure 6. Differential melting curves of nusinersen duplexes. The blank profile of the noncomplementary oligonucleotide (NSP) is shown as a dotted line.

3.3. Spectral Characteristics

Four definite methods were employed for comprehensive characterization of nusinersen spectral properties: ultraviolet (UV) spectroscopy, Fourier-transform infrared (FTIR) spectroscopy, intrinsic fluorescence (IF) spectroscopy, and circular dichroism (CD) spectroscopy. The UV absorption spectra of both the generic and reference products ex-hibited a characteristic peak at 260 nm, indicative of nucleic acid content (Figure 7A). These spectra were utilized to determine precise concentrations via the molar extinction coefficient, which was considered important for subsequent evaluation of biological ac-tivity.

Figure 7B presents the circular dichroism (CD) spectra of nusinersen samples recorded at 20 °C. A visual comparison of GNR-100 data revealed that the molar ellipticities at the principal extrema (207, 222, 245, and 273 nm) were similar to those of the RLD. Such spectral congruence is indicative of a similar arrangement of phosphorothioate linkages within the nucleotide sequence and their corresponding stereochemical configuration [18,21].

The infrared spectra (IR) of GNR-100 and the RLD exhibited nearly superimposable absorption bands (Figure 7C); a detailed assignment of these features is provided in Table S2. Analysis of the positions of spectral minima also confirmed that the IR spectra were similar across the wavenumber range of 1800–650 cm^{-1}. Minor variations in peak intensi-ties may arise from differential hydration, attributable to the hygroscopic nature of the samples.

Although fluorescent quantum yields of nucleic acids at ambient temperature are ex-tremely low, contributing to their reputation as nonfluorescent molecules, the monomeric chromophores of 5-methyl cytidine may exhibit appreciable intrinsic fluorescence [22]. Upon excitation at 280 nm, the intrinsic fluorescence spectra of both drug products dis-played a single emission maximum at approximately 346 nm (Figure 7D). Statistical anal-ysis of mean lognormal spectral parameters revealed no significant differences between GNR-100 and reference nusinersen (Table S3).

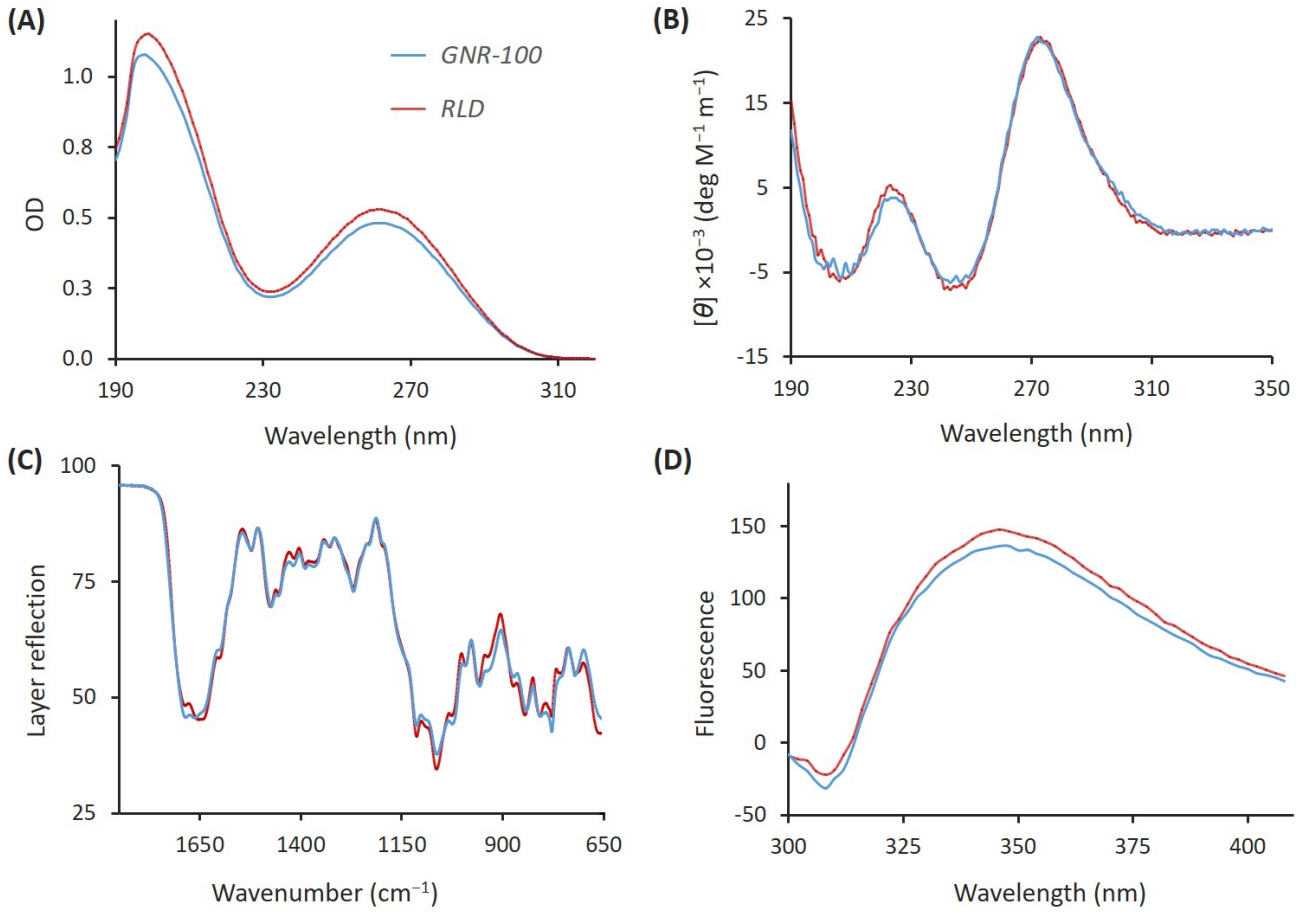

Figure 7. Spectral characterization of GNR-100 (blue) and the originator nusinersen (red): UV spectra (**A**); CD spectra (**B**); IR spectra (**C**); intrinsic fluorescence after 280 nm light excitation (**D**).

3.4. Biological Activity

SMN protein expression in nusinersen-treated fibroblasts was quantified as a percentage relative to the RLD batch, defined as 100% activity. Fibroblast samples were derived from three donors with spinal muscular atrophy. The findings confirmed that the enhancement of functionally active SMN protein synthesis across all GNR-100 samples remained within the established quality range (QR) of reference nusinersen (Figure 8A). The ability of nusinersen to promote transcription of full-length SMN2 mRNA, including exon 7, was separately evaluated using the same fibroblasts derived from patients with SMA. Primary fibroblasts harboring three copies of the SMN2 gene exhibited reduced transcription of the truncated SMN2Δ7 isoform. In contrast, transcription of the full-length SMN2 increased approximately twofold, with a consistent enhancement observed across all donors (Figure S6). Equivalence testing confirmed that GNR-100 and RLD were comparable in their ability to modulate SMN2 mRNA transcription levels (Figure 8B).

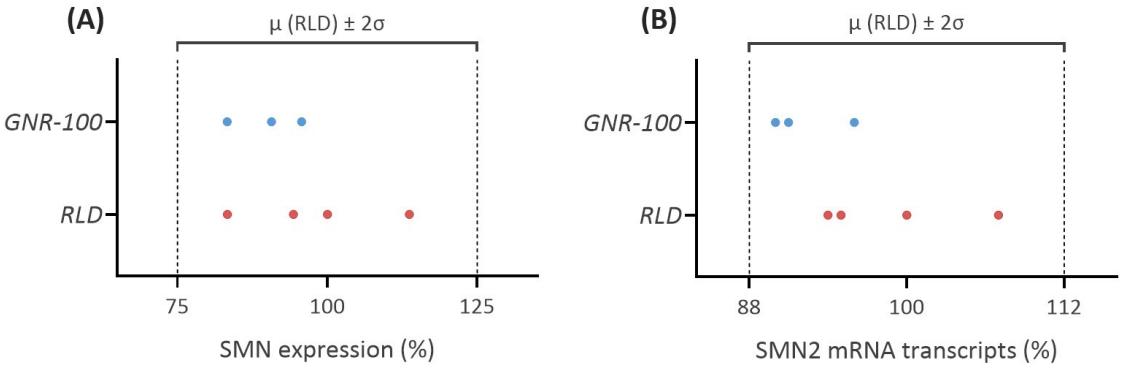

Figure 8. Biological activity after patient cell transfection with nusinersen. Fibroblast SMN expression (**A**) and levels of SMN2 mRNA transcripts (**B**).

3.5. Purity

Ion-pair HPLC-MS analysis provided the m/z ratios and relative abundances of related substances, enabling a direct comparison between generic and RLD products. This method confirmed the presence of impurities characteristic of chemically synthesized oligonucleotides, including raw materials as well as product- and process-related species (Table S4). The UV chromatograms as well as MS profiles of GNR-100 and RLD were sim-ilar, indicating comparable purity. For most impurities, levels in the generic nusinersen samples did not exceed those in the RLD (Figure S7, Table 1). Size exclusion chromatog-raphy was used to assess the molecular mass distribution of nusinersen and to estimate the content of aggregates, which are potential immunogenic agents. Both drug products demonstrated a prominent monomer peak at approximately 16 min, with monomer pu-rity exceeding 99.9% (Figure S8). Owing to its lower overall impurity content, the target oligonucleotide concentration in GNR-100 batches was 95–97%, as summarized in Table 1.

The sodium content in synthetic oligonucleotide samples was determined using ICP-OES, whereas the content of other elemental impurities was determined using ICP-MS. The purity assessment findings are summarized in Table 1.

4. Discussion

Drug products that replicate the structural and functional properties of original innovative molecules play a special role in modern drug development and production. Un-like generic drug products containing chemically synthesized low-molecular-weight com-pounds, biosimilars typically have a considerably more complex structure, resulting in high sensitivity to modifications and variations in the production process. Synthetic oli-gonucleotides combine the characteristics of both of the mentioned groups. Their produc-tion typically employs classical organic synthesis techniques [23]. However, the resulting molecules are quite complex, making oligonucleotides more similar to classical biopoly-mers such as peptides or glycoproteins [24].

The nusinersen-based drug product manufactured under the internal name of GNR-100 using automated solid-phase phosphoramidite technology is a synthetic 18-mer oli-goribonucleotide with modified phosphorothioate bonds and 2′-O-ester protected hy-droxyl atoms to ensure the drug products resistance to cellular nucleases (Figure S1).

Comprehensive physicochemical, structural, and biological studies were carried out using modern analytical procedures during a head-to-head comparative investigation of industrial batches of the proposed generic drug product GNR-100 versus RLD. To better

understand the characteristics and properties of the active ingredient of the durg product, a set of orthogonal methods was employed.

The findings of ion-pair HPLC-MS, including in tandem fragmentation mode, confirmed the identity of the primary structure as matching the expected theoretical sequence. The fine chemical structure of nusinersen was further characterized by nuclear magnetic resonance spectroscopy on $_{1}$H, $_{13}$C, and $_{31}$P nuclei. However, it was impossible to determine the complete chemical structure using 2D NMR. The comparable spectral and chromatographic profiles of both drug products, similar stability to phosphodiesterases, close melting points of the complementary duplexes, and comparable biological activity characteristics further confirm the identity of nusinersen properties in GNR-100 and the reference product.

According to recent knowledge, the stereochemical purity is able to affect the pharmacological properties of oligonucleotide-based drug products. Therefore, the synthesis and selection of the most active stereoisoforms are relevant [25–28]. The original nusinersen-based drug product represents a racemic mixture containing 2_{17} (131,072) structural isoforms, with variations in the starting components or synthesis conditions obviously capable of modulating the final distribution. Randomized distribution itself can provide a positive therapeutic effect; however, the permissible limits of such variations must be assessed for their impact on the efficacy and safety of oligonucleotide molecules [29,30]. Regarding the generic product GNR-100, it is reasonable to rule out direct changes in the diastereomeric composition relative to the originator nusinersen. The technologies used for the synthesis of both products do not allow for full control over the stereochem-ical variations of phosphorothioate (S_P) linkages. Therefore, the similarity of the diastere-omeric composition was assessed using a combination of analytical and functional meth-ods. This is supported by comparable molar ellipticity indices and melting points of the complementary duplexes. The $_{31}$P NMR spectra of all samples featured two distinct signal clusters: one for the R_P diastereomers downfield at 56.7–60.0 ppm and another for the S_P diastereomers at 53.0–56.7 ppm [31]. When comparing the relative signal intensities, a similar R_P/S_P diastereomeric ratio was observed across the studied drug products (Table 1).

Both nusinersen-based drug products equally increased the production of functional SMN protein in fibroblasts derived from patients with severe SMA. The products also exhibited similar resistance to nuclease degradation. Both were confirmed to effectively activate the transcription of full-length SMN2 mRNAs, including exon 7.

The related substance and target oligonucleotide impurity profile of the biosimilar candidate matches that of RLD. Furthermore, the levels of most identified GNR-100 impurities do not exceed those in the reference nusinersen, which may enhance its safety profile [32]. The generic drug product is generally free of heavy elements that could accu-mulate from industrial equipment or reagents. Traces of boron, iron, and silicon were de-tected, but their concentrations are within permissible limits [33].

5. Conclusions

A match or slight difference in CQA of the proposed drug product GNR-100 indicates its similarity to the original product. Based on the findings of the comparative study, the structure, diastereomeric composition, and chemical purity resulting in splice-switching activity of the active ingredient of generic nusinersen and the originator RLD are compa-rable in terms of quality, safety, and efficacy. Based on clinical trial data demonstrating favorable safety and efficacy profiles, the pharmaceutical product received regulatory ap-proval for clinical use in the Russian Federation in 2024, followed by authorization in Ka-zakhstan in 2025, thus further confirming its bioequivalence.

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Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/pharmaceutics18020178/s1, Figure S1: Chemical formula of nusinersen sodium salt; Figure S2: Spectrum $_1$H NMR of GNR-100; Figure S3: Overlay of the $_1$H-$_{13}$C HSQC NMR spectra for the drug product GNR-100 and the reference nusinersen; Figure S4: Spectrum $_1$H-$_1$H TOCSY NMR of GNR-100; Figure S5: GNR-100, RLD, and oligonucleotide CTRL electropherograms before (A) and afterPDE I (B) or PDE II (C) phosphodiesterase treatment; Figure S6: SMN2 FL and SMN2 Δ7 gene transcription in SMA patientsfibroblasts under GNR-100 promotion; Figure S7: An example of nusinersen ion-pairing HPLC-RP profiles; Figure S8: An example of nusinersen SEC profiles; Table S1: Fragment ions used for nusinersen structure evaluation; Table S2: IR spectroscopy data; Table S3: Lognormal parameters of IF spectra. Table S4: A list of potential product-related impurities of nusinersen.

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Abbreviations

The following abbreviations are used in this manuscript:

AUC Analytical Ultracentrifugation
CD Circular Dichroism
CQA Critical Quality Attribute
DMT 4,4-Dimethoxytrityl
FDA U.S. Food and Drug Administration
FTIR Fourier-Transform Infrared Spectroscopy
HFIP 1,1,1,3,3,3-Hexafluoroisopropanol
HPLC High Performance Liquid Chromatography
HPLC-RP Reversed-Phase High Performance Liquid Chromatography
ICP Inductively Coupled Plasma
IF Intrinsic Fluorescence
JSC Joint Stock Company
LC-MS Liquid Chromatography–Mass Spectrometry
NMR Nuclear Magnetic Resonance
OD Optical Density
PBS Phosphate-Buffered Saline
PDE Phosphodiesterase
QR Quality Range
RLD Reference Listed Drug
RNA Ribonucleic Acid

	RT PCR	Real-Time Polymerase Chain Reaction
	SD	Standard Deviation
SMA		Spinal Muscular Atrophy
	SMN2	Survival Motor Neuron 2 Gene
TEA		Triethylamine

Appendix A

Appendix A.1. Detailed Experimental Procedures

Appendix A.1.1. Synthetic Oligonucleotides

Short 2′-oligodeoxyribonucleotide primers were synthesized by Evrogen using an automated phosphoramidite method (Table A1).

Table A1. Synthetic oligonucleotides used in the study.

Name	Sequence (5′→3′)
ANT	CCAGCATTATGAAAGTGA
NSP	CAGCCACTGGTAACAGGAT
SMN2 FL SMN2 P	GCTCACATTCCTTAAATTAAGG
SMN2 Δ7 GAPDH 1 GAPDH 2 HEX	TCCAGATCTGTCTGATCGTT
ROX	TGGCTATCATACTGGCTATTATA
	CAACGGATTTGGTCGTATT
	TGATGGCAACAATATCCACT
	(HEX)-
	CTGGCATAGAGCAGCACTAAATGA
	(BHQ-2) (ROX)-CGCCTGGTCACCAG
	(BHQ-2)

Appendix A.1.2. Scanning and Tandem Chromatography–Mass Spectrometry

Prior to analysis, nusinersen samples were purified from low-molecular-weight com-ponents by ultrafiltration on an Amicon Ultra-0.5 (3 kDa) device, using water as the elu-ent. Samples with a concentration of approximately 0.5 mg/mL were analyzed by ion-pair HPLC-RP using an ACQUITY Premier Oligonucleotide BEH C18 column, 2.1 × 150 mm, 1.7 μm (Waters, Milford, MA, USA), and an Infinity II Bio 1260 system (Agilent Technol-ogies, Santa Clara, CA, USA) coupled with a UV/visible detector and an X500B QTOF mass spectrometer (Sciex, Framingham, MA, USA). Separation was performed at 85 °C at a flow rate of 0.2 mL/min using a methanol gradient in the mobile phase (15 mM TEA in water, 0.4 M HFIP). The gradient was as follows: 0 min, 73%; 12 min, 58%; 16 min, 10%; 17 min, 10%; 17.01 min, 73%; 32 min, 73%. The absorption signal was recorded at 260 nm; mass spectra were measured by negative ionization, with frontal and tandem data-de-pendent recording. The input capillary voltage was +4.5 kV; the drying, focusing, and curtain gas pressures were 70, 70, and 35 psi, respectively; the CAD gas pressure was 7 units; and the source temperature was 350 °C. In the frontal mode, the scanning range was 600–4000 m/z, with a declustering potential of 60 V and a fixed collision energy of 7 V. The nusinersen $(M\text{-}7H)^{7-}$ ion was fragmented in a tandem data-dependent mode with an m/z of 1017.03, collision energy of 28 V, ion accumulation time of 0.8 s, and QJet amplitude of 200 V; the scan range was 100–3000 m/z. Experimental data were processed using Sciex OS v.3.0 (Sciex) and Byos v.5.2.31 (Protein Metrics, Inc., Boston, MA, USA) software.

Appendix A.1.3. NMR Spectroscopy

Prior to analysis, lyophilized samples were dissolved in 0.5 mL of deionized water; 20 μL of D_2O was then added, and the samples were transferred to 5 mm NMR tubes (Wilmad, Vineland, NJ, USA). The ^{1}H and ^{13}C spectra were obtained using an Avance-II 700 spectrometer (Bruker, Billerica, MA, USA) with a CryoProbe triple resonance probe with cooled coils and a preamplifier for the ^{1}H and ^{13}C channels. The spectra were

recorded at 45 °C. Data were collected, processed, and analyzed using Bruker Topspin v.3.2 software. The 90-degree pulse duration was calibrated for each sample at the beginning of measurement, using stroboscopic nutation for water proton signals [34]. The chemical shifts of $_{1}$H signals were calibrated relative to the water protons, whose signal was set at 4.7 ppm. The chemical shifts of $_{13}$C heteronuclei were calibrated indirectly according to IUPAC guidelines [35].

$_{31}$P NMR spectra were recorded using an Avance-III 400 UltraShield Plus spectrometer (Bruker, Billerica, MA, USA) with a resonance frequency of 163 MHz for $_{31}$P nuclei at 23 °C. A 0.1 g/mL solution of synthetic oligonucleotide in deuterated water was prepared and placed in an ampoule. The spectrum of 8% phosphoric acid solution in D_2O (0.0 ppm chemical shift) was recorded as an external standard.

The main measurement parameters are summarized in Table A2.

Table A2. NMR spectra registration parameters.

Mode	Parameters
1D $_{1}$H	1D $_{1}$H with solvent suppression, spectral width 25 ppm, resolution 1.1 Hz per point (32,768 FID real and imaginary data points), 10.9 or 1.9 sec relaxation delay (including AQ). Programmed im-pulse exponential apodization.
1D $_{13}$C	1D $_{13}$C proton-decoupled, spectral width 205 ppm, resolution 0.9 Hz per point (65,536 FID real and
1D $_{31}$P	1D $_{31}$P, spectral width 406 ppm, resolution 1.2 Hz per point (65,782 FID real and imaginary data points).
2D $_{13}$C-HSQC	2D $_{13}$C-HSQC with improved sensitivity, phase-sensitive with selection of magnetization gradi-ents. Spectral
2D 1H-1H TOCSY	2D 1H-1H TOCSY, 80 ms mixing time, MLEV mixing sequence, solvent suppression by excitation sculpting. Spectral width 25 · 10 ppm ($_{1}$H · $_{13}$C), 16,384 · 1024 data points.

Chemical shifts were compared in accordance with the principles described in [36]. The integral ratio of diastereomers (R_P/S_P) in phosphorothioate oligonucleotides was determined based on the differentiation of signal clusters in characteristic spectral regions according to [18].

Appendix A.1.4. Ion-Pair HPLC-RP (UV)

Samples of the compared drug products were analyzed by HPLC-RP without dilution. Separation was performed on an XBridge BEH C18 column (4.6 × 250 mm, Waters, Milford, MA, USA) at 50 °C with a flow rate of 0.25 mL/min. A gradient elution was em-ployed, increasing the acetonitrile content from 35 to 60% over 22 min within a mobile phase consisting of 5 mM aqueous tributylammonium acetate and 1 μM EDTA. The injec-tion volume was 20 μL, and detection was carried out at 260 nm.

Appendix A.1.5. Size Exclusion Chromatography

Samples were diluted to a nusinersen concentration of approximately 1.0 mg/mL and analyzed by SEC. Separation was performed on an Acquity UPLC Protein BEH 200 col-umn (4.6 × 150 mm, Waters, Milford, MA, USA) at 30 °C in isocratic mode, using a mobile phase of 20 mM Na-phosphate buffer (pH 7.0) containing 0.3 M NaCl at a flow rate of 0.15 mL/min. The injection volume was 2.5 μL, and detection was performed at 260 nm.

Appendix A.1.6. Nuclease Resistance

Samples of the compared drug products were tested for resistance to snake venom phosphodiesterase (PDE I, Sigma Aldrich, St. Louis, MO, USA, P3243) and bovine spleen

phosphodiesterase (PDE II, Sigma Aldrich, St. Louis, MO, USA, P9041). Reactions contained approximately 1.8 μg of oligonucleotide in 10 μL of the appropriate buffer (PDE I, 50 mM Tris-HCl, pH 7.2, 10 mM $MgCl_2$; or PDE II, 50 mM Na-acetate, pH 6.5, 10 mM $MgCl_2$) with 0.5 mU of enzyme and were incubated at 37 °C for 1 h. Enzymes were heat-inactivated at 100 °C for 2 min, and the mixtures were subsequently analyzed.

Oligonucleotide samples (0.75–2.0 μg) were mixed with an equal volume of 2X RNA Gel Loading Dye (Thermo Scientific, Waltham, MA, USA, R0641), heated at 90 °C for 2 min, and chilled on ice. Samples were analyzed by electrophoresis in a 20% polyacryla-mide gel (20 × 20 cm, 0.75 mm thick; AppliChem, A0385,0500) containing 7 M urea, using TBE running buffer (Thermo Fisher Scientific, Waltham, MA, USA, J62788.K3). The gel was pre-equilibrated at 1 V/cm for 30 min prior to sample loading. Electrophoresis was performed at 10 V/cm for 1 h. Following separation, the gel was fixed in a solution of 50% methanol and 10% acetic acid for 20 min and washed three times (3 min each) with deion-ized water. For detection, the gel was stained by incubation for 30 min under slow agita-tion in TBE buffer containing 0.01% SYBR™ Safe DNA Gel Stain (Thermo Fisher Scientific, Waltham, MA, USA, S33102) and 0.01% ethidium bromide (Thermo Fisher Scientific, Wal-tham, MA, USA, 15585011). Fluorescence imaging was performed using the Im-ageQuant™ 800 system (Cytiva, Marlborough, MA, USA) with excitation at 360 nm and emission recorded at 605 nm.

Appendix A.1.7. Melting Point of the Duplex with the Complementary Strand

The melting point (Tm) of the duplex was determined using an ANT oligonucleotide probe containing a complementary sequence. A 25 μL aliquot of the test sample was com-bined with 100 μM ANT oligonucleotide in reaction buffer (20 mM Tris-HCl, pH 8.4, 60 mM KCl, 1 mM $MgCl_2$) in a 96-well microplate. A non-specific oligonucleotide (NSP) was used as a reference. Subsequently, 1 μL of a 0.5% SYBR Green I dye solution in DMSO (Evrogen, Moscow, Russia, PB025M) was added to each well. Temperature and fluores-cence were monitored in real time using a CFX96 Touch™ Real-Time PCR Detection Sys-tem (Bio-Rad, Hercules, CA, USA). The mixtures were heated at 95 °C for 10 min. Next, the complementary chains were annealed by cooling to 20 °C for 10 min to allow duplex formation. A gradual temperature increase from 20 °C to 95 °C at a rate of 0.1 °C/s was monitored by the fluorescence signal at 521 nm with excitation at 497 nm throughout the melting phase to generate the melting curve.

Appendix A.1.8. UV Spectroscopy

UV absorption spectra were recorded using a dual-beam Cary 100 spectrophotometer (Varian Inc., Palo Alto, CA, USA). Scans were performed from 190 to 350 nm, with a 2 nm slit width and an averaging time of 0.1 s. The absorption spectrum of the placebo was subtracted from the spectra of the oligonucleotide-based drug products. Oligonucleotide concentration was calculated using the molar extinction coefficient at 260 nm (144,807 $M^{-1}cm^{-1}$). The weight concentration was subsequently determined using the relative mo-lecular mass of nusinersen sodium salt (7501 g/mol).

Appendix A.1.9. Fourier-Transform Infrared Spectroscopy

Each sample was dissolved in 1 mL of water and dialyzed at 4 °C in a single container against 0.5 L of water using dialysis wells with a 3.5 kDa cutoff: once for 15 h, followed by three exchanges for 2 h each. Subsequently, 0.7 mL aliquots were frozen and lyophilized. FTIR spectra of the lyophilized samples were recorded at 29.5 °C using a Nicolet 6700 spectrometer (Thermo Scientific Waltham, MA, USA) equipped with a cooled MCT detec-tor. Measurements were performed on a diamond crystal in attenuated total reflectance

(ATR) mode with an optical path length of 2 μm. Spectra were acquired over the wavenumber range of 650–4000 cm^{-1} by averaging 256 scans.

Appendix A.1.10. CD Spectroscopy

Circular dichroism spectra were recorded on a J-810 spectropolarimeter (JASCO Inc., Easton, MD, USA) using a 10 × 1 mm quartz cuvette (Hellma, Mullheim, Germany). Meas-urements were performed at 20 °C over a wavelength range of 195–350 nm, with a slit width of 2 nm and a data averaging time of 2 s. The oligonucleotide concentration was approximately 15 μM for all measurements.

Appendix A.1.11. Intrinsic Fluorescence Spectroscopy

Fluorescence spectra were recorded on a Cary Eclipse spectrofluorimeter (Varian Inc., Palo Alto, CA, USA) at 20 °C using a 1 cm quartz cuvette. Measurements were performed in the emission wavelength range of 300–410 nm following excitation at 280 nm, with a data averaging time of 1 s. The excitation and emission slit widths were 10 and 5 nm, respectively. The oligonucleotide concentration was approximately 0.9–1.0 μM.

Appendix A.1.12. ICP-OES

To determine the sodium counterion content, a stock solution of the drug product containing 10–100 μg/mL nusinersen in 1 M nitric acid was prepared. Sodium analysis was performed by inductively coupled plasma emission spectrometry using an Optima 5300DV spectrometer (Perkin Elmer, Shelton, CT, USA). The instrument was operated with a generator frequency of 40 MHz, an output power of 1.3 kW, and an argon flow rate of 15 L/min for plasma formation. Auxiliary and cooling flow rates were each 0.5 L/min, and the sample introduction rate was 0.85 mL/min. The plasma was allowed to stabilize for 30 min prior to analysis.

Appendix A.1.13. ICP-MS

To determine heavy metals, a sample solution containing 10–60 μg/mL of the drug product in 1 M nitric acid was prepared for analysis. Measurements were performed using an Elan DRC-e inductively coupled plasma emission spectrometer (Perkin Elmer, Shelton, CT, USA), with the following operating parameters: generator frequency of 40 MHz, out-put RF power of 1.1 kW, plasma gas (Ar) flow of 15 L/min, and auxiliary gas (Ar) flow of 0.9 L/min. The sample introduction rate was 0.85 mL/min.

Appendix A.1.14. Sedimentation Analysis

Analytical ultracentrifugation of the drug products was performed using an Optima-AUC centrifuge (Beckman Coulter, Brea, CA, USA) equipped with an An-60 Ti rotor and 12 mm A1-Epon double-sector cells. Samples were prepared in aqueous solution at nusinersen concentrations of 30 μg/mL and 60 μg/mL. Following a 70-min equilibration at 20 °C, centrifugation was conducted at 260,000× *g*. Absorbance scans were recorded at 260 nm at 400 s intervals over 5 h, using water as the reference. Data were processed using the maximum entropy regularization method.

Appendix A.1.15. SMN Protein Biosynthesis

Primary fibroblasts from three SMA donors (250,000 cells per well) were seeded in 2 mL of DMEM/F12 medium (PanEco, Moscow, Russia) supplemented with 15% fetal bovine serum, 2 mM L-glutamine, 60 μg/mL penicillin, and 100 μg/mL streptomycin, and pre-cultured for 18 h at 37 °C. For transfection, a mixture of 3 μL of the drug product and 747 μL Opti-MEM™ I (Thermo Scientific, Waltham, MA, USA) was combined with a mixture of 18 μL of Lipofectamine™ 2000 (Thermo Scientific, Waltham, MA, USA) and 732 μL

Opti-MEM™ I. This transfection complex was incubated at room temperature for 30 min. A 0.5 mL aliquot of the resulting mixture was then added to each well of pre-seeded fibroblast and incubated at 37 °C under 5% CO_2 with gentle agitation for 48 h. After incubation, cells were washed three times with 5 mL of DMEM/F12 medium and twice with PBS. Cells were lysed by adding 0.5 mL of deionized water to each well and subjected to one freeze-thaw cycle. The resulting lysates were clarified by centrifugation at 1000 rpm for 5 min.

The total soluble protein content of the clarified lysates was quantified using a Pierce BCA Protein Assay Kit (Thermo Scientific, Waltham, MA, USA, 23227). SMN protein ex-pression was measured with a specific ELISA kit (Enzo Life Sciences, Farmingdale, NY, USA, ADI-900-209). The specific SMN content was calculated as nanograms of SMN per mg of total protein, normalized to the BCA assay results.

Appendix A.1.16. SMN Gene Expression

Fibroblasts transfected with nusinersen were used to quantify full-length SMN mRNA transcripts (SMN1 + SMN2). Cells from three SMA donors harboring three or four copies of the SMN2 gene were treated with nusinersen at concentrations of 8, 40, and 80 nM. Total RNA was isolated using an AllPrep DNA/RNA/miRNA Universal Kit (QI-AGEN, Hilden, Germany). RNA concentration was assessed by spectrophotometry on a NanoDrop 1000 system (Thermo Scientific, Waltham, MA, USA).

Relative mRNA expression was determined by real-time qPCR using a CFX96 Touch™ system (Bio-Rad, Hercules, CA, USA). Each 20 μL reaction contained 10 μL of isolated mRNA (diluted to 2.5 ng/μL in nuclease-free water), 4 μL of 5× Genta Single-tube RT-PCR master mix (SynGen, Sacramento, CA, USA), 4 μL of nuclease-free water, and 2 μL of the primer/probe mixture. For amplification of the reference housekeeping gene (GAPDH), a ROX probe (0.75 μM) and GAPDH 1 and GAPDH 2 oligonucleotides (1 μM each) were used. For amplification of the SMN2 region, a HEX probe (1.5 μM) and SMN2 P oligonucleotide (4 μM) were used. The SMN2 FL oligonucleotide (4 μM) was used as the second primer for the full-length protein, and the SMN2 Δ7 oligonucleotide (4 μM) was used for SMN2 with deletion in exon 7.

The thermal cycling protocol consisted of cDNA synthesis at 55 °C for 30 min, reverse transcriptase inactivation at 95 °C for 15 min, followed by multi-cycle PCR amplification with denaturation at 95 °C for 15 s and annealing/extension at 60 °C for 1 min.

The relative expression of each SMN2 transcript isoform was calculated using the $2^{-\Delta Ct}$ method, where ΔCt represents the difference in threshold cycles between the target (SMN2) and reference (GAPDH) genes. Fold-change in expression was determined as the ratio of these values in nusinersen-treated versus untreated cells.

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